INNER JOIN:

```
lab_14=# SELECT c.name, c.email, o.order_date, o.total_amount
lab_14-# FROM customers c
lab_14-# INNER JOIN orders o ON c.customer_id = o.customer_id;
       name       |            email             | order_date | total_amount
------------------+------------------------------+------------+--------------
 Aigerim Sadykova | aigerim.sadykova@example.com | 2023-11-15 |       176.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-02-10 |       850.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-03-22 |        95.00
 Timur Asanov     | timur.asanov@example.com     | 2023-12-05 |       246.00
 Timur Asanov     | timur.asanov@example.com     | 2024-01-18 |       565.00
 Nurlan Bekov     | nurlan.bekov@example.com     | 2024-02-28 |        75.50
 Dana Karimova    | dana.karimova@example.com    | 2024-03-03 |       875.00
 Dana Karimova    | dana.karimova@example.com    | 2024-04-14 |       420.00
 Emil Joldoshev   | emil.joldoshev@example.com   | 2023-10-30 |        60.00
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-09 |       595.50
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-21 |       165.00
 John Carter      | john.carter@example.com      | 2024-06-02 |      1105.00
 Timur Asanov     | timur.asanov@example.com     | 2024-06-15 |       175.00
(13 rows)
```

LEFT JOIN (LEFT OUTER JOIN):

```
lab_14=# SELECT c.name, c.email, o.order_date, o.total_amount
lab_14-# FROM customers c
lab_14-# LEFT JOIN orders o ON c.customer_id = o.customer_id;
       name       |            email             | order_date | total_amount
------------------+------------------------------+------------+--------------
 Aigerim Sadykova | aigerim.sadykova@example.com | 2023-11-15 |       176.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-02-10 |       850.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-03-22 |        95.00
 Timur Asanov     | timur.asanov@example.com     | 2023-12-05 |       246.00
 Timur Asanov     | timur.asanov@example.com     | 2024-01-18 |       565.00
 Nurlan Bekov     | nurlan.bekov@example.com     | 2024-02-28 |        75.50
 Dana Karimova    | dana.karimova@example.com    | 2024-03-03 |       875.00
 Dana Karimova    | dana.karimova@example.com    | 2024-04-14 |       420.00
 Emil Joldoshev   | emil.joldoshev@example.com   | 2023-10-30 |        60.00
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-09 |       595.50
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-21 |       165.00
 John Carter      | john.carter@example.com      | 2024-06-02 |      1105.00
 Timur Asanov     | timur.asanov@example.com     | 2024-06-15 |       175.00
 Meerim Bekova    | meerim.bekova@example.com    |            |
 Maria Lopez      | maria.lopez@example.com      |            |
 Bakyt Mamatov    | bakyt.mamatov@example.com    |            |
(16 rows)
```

RIGHT JOIN (RIGHT OUTER JOIN):

```
lab_14=# SELECT c.name, c.email, o.order_date, o.total_amount
lab_14-# FROM customers c
lab_14-# RIGHT JOIN orders o ON c.customer_id = o.customer_id;
       name       |            email             | order_date | total_amount
------------------+------------------------------+------------+--------------
 Aigerim Sadykova | aigerim.sadykova@example.com | 2023-11-15 |       176.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-02-10 |       850.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-03-22 |        95.00
 Timur Asanov     | timur.asanov@example.com     | 2023-12-05 |       246.00
 Timur Asanov     | timur.asanov@example.com     | 2024-01-18 |       565.00
 Nurlan Bekov     | nurlan.bekov@example.com     | 2024-02-28 |        75.50
 Dana Karimova    | dana.karimova@example.com    | 2024-03-03 |       875.00
 Dana Karimova    | dana.karimova@example.com    | 2024-04-14 |       420.00
 Emil Joldoshev   | emil.joldoshev@example.com   | 2023-10-30 |        60.00
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-09 |       595.50
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-21 |       165.00
 John Carter      | john.carter@example.com      | 2024-06-02 |      1105.00
 Timur Asanov     | timur.asanov@example.com     | 2024-06-15 |       175.00
                  |                              | 2024-07-01 |        95.00
(14 rows)
```

FULL OUTER JOIN:

```
lab_14=# SELECT c.name, c.email, o.order_date, o.total_amount
lab_14-# FROM customers c
lab_14-# FULL OUTER JOIN orders o ON c.customer_id = o.customer_id;
       name       |            email             | order_date | total_amount
------------------+------------------------------+------------+--------------
 Aigerim Sadykova | aigerim.sadykova@example.com | 2023-11-15 |       176.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-02-10 |       850.00
 Aigerim Sadykova | aigerim.sadykova@example.com | 2024-03-22 |        95.00
 Timur Asanov     | timur.asanov@example.com     | 2023-12-05 |       246.00
 Timur Asanov     | timur.asanov@example.com     | 2024-01-18 |       565.00
 Nurlan Bekov     | nurlan.bekov@example.com     | 2024-02-28 |        75.50
 Dana Karimova    | dana.karimova@example.com    | 2024-03-03 |       875.00
 Dana Karimova    | dana.karimova@example.com    | 2024-04-14 |       420.00
 Emil Joldoshev   | emil.joldoshev@example.com   | 2023-10-30 |        60.00
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-09 |       595.50
 Aliya Omarova    | aliya.omarova@example.com    | 2024-05-21 |       165.00
 John Carter      | john.carter@example.com      | 2024-06-02 |      1105.00
 Timur Asanov     | timur.asanov@example.com     | 2024-06-15 |       175.00
                  |                              | 2024-07-01 |        95.00
 Meerim Bekova    | meerim.bekova@example.com    |            |
 Maria Lopez      | maria.lopez@example.com      |            |
 Bakyt Mamatov    | bakyt.mamatov@example.com    |            |
(17 rows)
```

CROSS JOIN:

```
lab_14=# SELECT c.name, p.product_name
lab_14-# FROM customers c
lab_14-# CROSS JOIN products p;
       name        |  product_name
-------------------+-----------------
 Aigerim Sadykova | Laptop
 Timur Asanov     | Laptop
 Nurlan Bekov     | Laptop
 Dana Karimova    | Laptop
 Emil Joldoshev   | Laptop
 Aliya Omarova    | Laptop
 John Carter      | Laptop
 Maria Lopez      | Laptop
 Bakyt Mamatov    | Laptop
 Meerim Bekova    | Laptop
 Aigerim Sadykova | Smartphone
 Timur Asanov     | Smartphone
 Nurlan Bekov     | Smartphone
 Dana Karimova    | Smartphone
 Emil Joldoshev   | Smartphone
 Aliya Omarova    | Smartphone
 John Carter      | Smartphone
 Maria Lopez      | Smartphone
 Bakyt Mamatov    | Smartphone
 Meerim Bekova    | Smartphone
 Aigerim Sadykova | Headphones
 Timur Asanov     | Headphones
 Nurlan Bekov     | Headphones
 Dana Karimova    | Headphones
 Emil Joldoshev   | Headphones
 Aliya Omarova    | Headphones
 John Carter      | Headphones
 Maria Lopez      | Headphones
 Bakyt Mamatov    | Headphones
```

MULTIPLE TABLE JOINS:

```
lab_14=# SELECT c.name, o.order_date, oi.quantity, p.product_name, p.price
lab_14-# FROM customers c
lab_14-# INNER JOIN orders o ON c.customer_id = o.customer_id
lab_14-# INNER JOIN order_items oi ON o.order_id = oi.order_id
lab_14-# INNER JOIN products p ON oi.product_id = p.product_id;
      name       | order_date | quantity |  product_name    | price
-----------------+------------+----------+------------------+--------
 Aigerim Sadykova | 2023-11-15 |        2 | Headphones       |  75.50
 Aigerim Sadykova | 2023-11-15 |        1 | Mouse            |  25.00
 Aigerim Sadykova | 2024-02-10 |        1 | Laptop           | 850.00
 Aigerim Sadykova | 2024-03-22 |        1 | Keyboard         |  45.00
 Aigerim Sadykova | 2024-03-22 |        2 | Mouse            |  25.00
 Timur Asanov     | 2023-12-05 |        1 | Monitor          | 210.00
 Timur Asanov     | 2023-12-05 |        3 | USB Flash Drive  |  12.00
 Timur Asanov     | 2024-01-18 |        1 | Smartphone       | 520.00
 Timur Asanov     | 2024-01-18 |        1 | Keyboard         |  45.00
 Nurlan Bekov     | 2024-02-28 |        1 | Headphones       |  75.50
 Dana Karimova    | 2024-03-03 |        1 | Laptop           | 850.00
 Dana Karimova    | 2024-03-03 |        1 | Mouse            |  25.00
 Dana Karimova    | 2024-04-14 |        2 | Monitor          | 210.00
 Emil Joldoshev   | 2023-10-30 |        5 | USB Flash Drive  |  12.00
 Aliya Omarova    | 2024-05-09 |        1 | Smartphone       | 520.00
 Aliya Omarova    | 2024-05-09 |        1 | Headphones       |  75.50
 Aliya Omarova    | 2024-05-21 |        2 | Keyboard         |  45.00
 Aliya Omarova    | 2024-05-21 |        3 | Mouse            |  25.00
 John Carter      | 2024-06-02 |        1 | Laptop           | 850.00
 John Carter      | 2024-06-02 |        1 | Monitor          | 210.00
 John Carter      | 2024-06-02 |        1 | Keyboard         |  45.00
 Timur Asanov     | 2024-06-15 |        2 | Headphones       |  75.50
 Timur Asanov     | 2024-06-15 |        2 | USB Flash Drive  |  12.00
(23 rows)
```

SELF JOINS:

```
lab_14=# SELECT e1.name AS employee, e2.name AS manager
lab_14-# FROM employees e1
lab_14-# LEFT JOIN employees e2 ON e1.manager_id = e2.employee_id;
        employee         |      manager
-------------------------+-------------------
 Altynai Sultanova       |
 Bakyt Ismailov          | Altynai Sultanova
 Chyngyz Tokoev          | Altynai Sultanova
 Dinara Asanova          | Bakyt Ismailov
 Erlan Kadyrov           | Bakyt Ismailov
 Fatima Zhunusova        | Chyngyz Tokoev
 Gulnara Abdyrakhmanova  | Chyngyz Tokoev
 Hasan Toktogulov        | Dinara Asanova
 Irina Petrova           | Dinara Asanova
 Jamilya Orozova         | Erlan Kadyrov
(10 rows)
```

JOIN WITH CONDITIONS:

```
lab_14=# SELECT c.name, o.order_date, o.total_amount
lab_14-# FROM customers c
lab_14-# INNER JOIN orders o ON c.customer_id = o.customer_id
lab_14-# WHERE o.order_date >= '2024-01-01'
lab_14-# AND o.total_amount > 100;
       name        | order_date | total_amount
-------------------+------------+--------------
 Aigerim Sadykova  | 2024-02-10 |       850.00
 Timur Asanov      | 2024-01-18 |       565.00
 Dana Karimova     | 2024-03-03 |       875.00
 Dana Karimova     | 2024-04-14 |       420.00
 Aliya Omarova     | 2024-05-09 |       595.50
 Aliya Omarova     | 2024-05-21 |       165.00
 John Carter       | 2024-06-02 |      1105.00
 Timur Asanov      | 2024-06-15 |       175.00
(8 rows)
```

ONE-TO-ONE RELATIONSHIPS:

```
lab_14=# SELECT u.username, up.first_name, up.last_name, up.phone
lab_14-# FROM users u
lab_14-# LEFT JOIN user_profiles up ON u.user_id = up.user_id;
 username  | first_name | last_name |     phone
-----------+------------+-----------+---------------
 johndoe   | John       | Doe       | +12025550101
 aigerim_s | Aigerim    | Sadykova  | +996700111222
 timur_a   | Timur      | Asanov    | +996555123456
 nurlan_b  | Nurlan     | Bekov     | +996777234567
 dana_k    | Dana       | Karimova  | +996700345678
 emil_j    | Emil       | Joldoshev | +996550456789
 aliya_o   |            |           |
 maria_l   |            |           |
(8 rows)
```

ONE-TO-MANY RELATIONSHIPS:

```
lab_14=# SELECT a.author_name, b.title, b.publication_year
lab_14-# FROM authors a
lab_14-# INNER JOIN books b ON a.author_id = b.author_id
lab_14-# ORDER BY a.author_name, b.publication_year;
    author_name    |         title          | publication_year
-------------------+------------------------+------------------
 Chingiz Aitmatov  | Jamila                 |             1958
 Chingiz Aitmatov  | The First Teacher      |             1962
 Fyodor Dostoevsky | Crime and Punishment   |             1866
 Fyodor Dostoevsky | The Idiot              |             1869
 Fyodor Dostoevsky | The Brothers Karamazov |             1880
 George Orwell     | Animal Farm            |             1945
 George Orwell     | Nineteen Eighty-Four   |             1949
 Jane Austen       | Pride and Prejudice    |             1813
 Leo Tolstoy       | War and Peace          |             1869
 Leo Tolstoy       | Anna Karenina          |             1878
(10 rows)
```

MANY-TO-MANY RELATIONSHIPS:

```
lab_14=# SELECT s.student_name, c.course_name, e.enrollment_date, e.grade
lab_14-# FROM students s
lab_14-# INNER JOIN enrollments e ON s.student_id = e.student_id
lab_14-# INNER JOIN courses c ON e.course_id = c.course_id
lab_14-# WHERE e.grade IS NOT NULL
lab_14-# ORDER BY s.student_name, c.course_name;
   student_name   |    course_name    | enrollment_date | grade
------------------+-------------------+-----------------+-------
 Aigerim Sadykova | Algorithms        | 2025-09-01      | B+
 Aigerim Sadykova | Databases         | 2025-09-01      | A
 Aigerim Sadykova | Statistics        | 2025-09-02      | A-
 Aliya Omarova    | Algorithms        | 2025-09-02      | A-
 Aliya Omarova    | Databases         | 2025-09-01      | A
 Dana Karimova    | Algorithms        | 2025-09-01      | A
 Emil Joldoshev   | Calculus          | 2025-09-03      | B
 Emil Joldoshev   | Operating Systems | 2025-09-04      | C+
 John Carter      | Operating Systems | 2025-09-04      | B+
 Nurlan Bekov     | Calculus          | 2025-09-03      | A
 Nurlan Bekov     | Databases         | 2025-09-02      | B
 Timur Asanov     | Calculus          | 2025-09-03      | B
 Timur Asanov     | Databases         | 2025-09-01      | A-
(13 rows)
```

INDEXES:

```
lab_14=# CREATE INDEX idx_orders_customer_id ON orders(customer_id);
CREATE INDEX
lab_14=# CREATE INDEX idx_order_items_order_id ON order_items(order_id);
CREATE INDEX
lab_14=# CREATE INDEX idx_order_items_product_id ON order_items(product_id);
CREATE INDEX
lab_14=# SELECT c.name, o.order_date
lab_14-# FROM customers c
lab_14-# INNER JOIN orders o ON c.customer_id = o.customer_id;
       name       | order_date
------------------+------------
 Aigerim Sadykova | 2024-03-22
 Aigerim Sadykova | 2024-02-10
 Aigerim Sadykova | 2023-11-15
 Timur Asanov     | 2024-06-15
 Timur Asanov     | 2024-01-18
 Timur Asanov     | 2023-12-05
 Nurlan Bekov     | 2024-02-28
 Dana Karimova    | 2024-04-14
 Dana Karimova    | 2024-03-03
 Emil Joldoshev   | 2023-10-30
 Aliya Omarova    | 2024-05-21
 Aliya Omarova    | 2024-05-09
 John Carter      | 2024-06-02
(13 rows)
```

TABLE ALIASES:

```
lab_14=# SELECT c.name, o.order_date
lab_14-# FROM customers c, orders o;
       name       | order_date
------------------+------------
 Aigerim Sadykova | 2023-11-15
 Aigerim Sadykova | 2024-02-10
 Aigerim Sadykova | 2024-03-22
 Aigerim Sadykova | 2023-12-05
 Aigerim Sadykova | 2024-01-18
 Aigerim Sadykova | 2024-02-28
 Aigerim Sadykova | 2024-03-03
 Aigerim Sadykova | 2024-04-14
 Aigerim Sadykova | 2023-10-30
 Aigerim Sadykova | 2024-05-09
 Aigerim Sadykova | 2024-05-21
 Aigerim Sadykova | 2024-06-02
 Aigerim Sadykova | 2024-06-15
 Aigerim Sadykova | 2024-07-01
 Timur Asanov     | 2023-11-15
 Timur Asanov     | 2024-02-10
 Timur Asanov     | 2024-03-22
 Timur Asanov     | 2023-12-05
 Timur Asanov     | 2024-01-18
 Timur Asanov     | 2024-02-28
 Timur Asanov     | 2024-03-03
 Timur Asanov     | 2024-04-14
 Timur Asanov     | 2023-10-30
 Timur Asanov     | 2024-05-09
 Timur Asanov     | 2024-05-21
 Timur Asanov     | 2024-06-02
 Timur Asanov     | 2024-06-15
 Timur Asanov     | 2024-07-01
 Nurlan Bekov     | 2023-11-15
 Nurlan Bekov     | 2024-02-10
 Nurlan Bekov     | 2024-03-22
```

CARTESIAN PRODUCTS:

```
lab_14=# SELECT c.name, o.order_date
lab_14-# FROM customers c
lab_14-# INNER JOIN orders o ON c.customer_id = o.customer_id;
       name       | order_date
------------------+------------
 Aigerim Sadykova | 2024-03-22
 Aigerim Sadykova | 2024-02-10
 Aigerim Sadykova | 2023-11-15
 Timur Asanov     | 2024-06-15
 Timur Asanov     | 2024-01-18
 Timur Asanov     | 2023-12-05
 Nurlan Bekov     | 2024-02-28
 Dana Karimova    | 2024-04-14
 Dana Karimova    | 2024-03-03
 Emil Joldoshev   | 2023-10-30
 Aliya Omarova    | 2024-05-21
 Aliya Omarova    | 2024-05-09
 John Carter      | 2024-06-02
(13 rows)
```

NULL VALUE CONFUSION:

```
lab_14=# SELECT c.name
lab_14-# FROM customers c
lab_14-# LEFT JOIN orders o ON c.customer_id = o.customer_id
lab_14-# WHERE o.customer_id IS NULL;
     name
---------------
 Maria Lopez
 Bakyt Mamatov
 Meerim Bekova
(3 rows)
```